Pipelines for DH307 PD Subtopics

Subtopic 4

Domain Adversarial Learning for Dataset and Language Invariant Parkinson's Disease Detection.

Literature and Dataset Links

Paper - https://www.mdpi.com/2306-5354/10/11/1316

Datasets
(1) NeuroVoz.zip
(2) MDVR-KCL and
(3) ItalianPD - https://iitbacin-my.sharepoint.com/:f:/r/personal/mayurakshi_mukherji_iitb_ac_in/Documents/Datasets/PD?csf=1&web=1&e=QicUHT

Preprocessing and Feature Extraction

Datasets

Datasets : Italian PD, NeuroVoz, MDVR-KCL
Tasks: Spontaneous speech, reading passage, words, and phrases

Basic audio pre-processing

i. Resample to 16 kHz
ii. Normalize amplitude for each file
iii. Remove noise files manually

Simple Segmentation

i. Fixed duration segments for sentence-level segmentation.
ii. Fixed duration segments for very short durations (w/o word or phoneme timestamps)

Alignment-based segmentation

i. Generate word-level and phoneme-level time-stamps by passing transcripts and audio through MFA or Wav2Vec2 Aligner.
ii. Segment based on time-stamps for phoneme and word clips.

Data splits

Create speaker-separated train, validation, and test splits.
i. Train set – PD + HC from multiple datasets
ii. Test set – PD + HC from each dataset kept separately

Feature extraction

Feature extraction for different types of segments
(1) phoneme-level,
(2) word-level,
(3) duration-level, and
(4) utterance-level

Features
i. Standard handcrafted features like – MFCCs, Jitter, Shimmer, ZCR, STE, F0, F1, F2, F3, Intensity, etc.
ii. Spectrograms

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Domain-adversarial learning

Input

Input to models

i. Extracted hand-crafted features
ii. Spectrograms

Architecture

ML models – DANN model consisting of 2 branches – i) for reducing classification loss of dataset/language classification, ii) for increasing classification accuracy of PD vs. HC.

Either or both branches can have any architecture – MLP or CNN or transformers etc.

A separate model for every feature and segmentation type combination.

For e.g. – 1 DANN models for *vowel phonemes* category using spectrogram feature.

Data

Training data

i. HC and PD from multiple datasets. Each tagged with the dataset and language they are associated with

Testing data

i. PD + HC data for each dataset tested separately

Testing

Results reported with metrics – precision, recall, accuracy, ROC-AUC for test set from each dataset for each segmentation level,

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